

Functions and Data Structures

Errors, exceptions and file handling

- ✔ **Video:** What are exceptions 2 min
- ✔ **Video:** Exception handling 4 min
- ✔ **Reading:** Exercise: Exceptions in Python 30 min
- ✔ **Reading:** Exceptions in Python - solution 10 min
- 📖 **Practice Quiz:** Exceptions in Python 3 questions
- ✔ **Video:** File handling in Python 5 min
- ✔ **Video:** Creating Files 5 min
- ✔ **Video:** Reading Files 5 min
- ✔ **Reading:** Storing file contents in data structures 20 min
- 📖 **Practice Quiz:** Read in data, store, manipulate and output new data to a file 5 questions
- 📖 **Programming Assignment:** Read in data, store, manipulate and output new data to a file 3h
- 📖 **Video:** Module summary: Basic Programming with Python 1 min
- 📖 **Quiz:** Module quiz: Basic Programming with Python 8 questions
- 📖 **Reading:** Additional resources 5 min

Storing file contents in data structures

Imagine you are trying to come up with a name for your new dog. You're really unsure of what you'd like to call the dog, so you decide to use your Python skills to help you decide.

You start by accessing a file with a shortlist of names you'd like to use for your new pet.

The file is named `petnames.txt`, and has the following content:

```
7  Buddy
8  Coco
9  Cooper
10 Duke
11 Dozer
12 Echo
13 Gizmo
14 Harley
15 Mac
16 Max
17 Milo
18 Oscar
19 Rex
20 Rocky
21 Rocket
22 Wolfie
```

Now that you have the file `petnames.txt`, you'd like to use this file inside your Python program to randomly pick a single pet name.

To do this, you'll need to have a Python file into which you'll be importing the `petnames.txt` file, as follows:

```
f = open("petnames.txt", "r")
```

The `open()` function reads in a file outside of the program itself.

The `open()` function accepts two parameters:

1. The path and the filename in the form of a string.
2. The import mode (the default being `"r"` - which means "read")

In the line above, I am importing the file at the root of the project. I am only specifying the filename, without the path. I am also using the default `"r"` mode to read in the file. I am saving the imported file into a variable named `f`.

Next, I'm going to add another variable, `f_content`, and I'm assigning to it the result of reading the `f` file.

On the third line, I'm printing the `f_content` variable.

The `print(f_content)` line returns the exact content of the file, as is:

```
10 Duke
11 Dozer
12 Echo
13 Gizmo
14 Harley
15 Mac
16 Max
17 Milo
18 Oscar
19 Rex
20 Rocky
21 Rocket
22 Wolfie
```

Now that I've confirmed that I'm successfully reading in the file, it would not be useful to keep printing out the file's contents, so I can comment out the `print(f_content)` line.

Additionally, I can get the `f_content` variable into a list. The string `"\n"` is used to split the text where a new line is found.

```
f_content_list = f_content.split("\n")
```

Now I'm ready to print the `f_content_list` variable, as follows:

```
print(f_content_list)
```

This time, the output is as follows:

```
1  ['Ace', 'Atlas', 'Bailey', 'Bear', 'Blaze', 'Boomer', 'Buddy', 'Coco', 'Cooper', 'Duke', 'Dozer', 'Echo', 'Gizmo', 'Harley', 'Mac', 'Max', 'Milo', 'Oscar', 'Rex', 'Rocky', 'Rocket', 'Wolfie']
```

Here's my complete code up to this point, with the redundant `print()` calls deleted.

```
1  f = open("petnames.txt", "r")
2  f_content = f.read()
3  f_content_list = f_content.split("\n")
```

Run
Reset

Now that I have all my potential pet names in a list, I can randomly pick a name from the `f_content_list` of names.

To do this, I'll need to import the random module at the top of my code: `import random`.

Now I can use the random module's `choice()` function: `random.choice()`.

The `choice()` function accepts a sequence parameter. A list is one of its accepted values. This means that I can now add another line of code at the very bottom of my program:

```
print(random.choice(f_content_list))
```

Running the code now will output a random pet name. The first time I ran it, I got the name "Milo", and the second time I ran it, I got the name "Dozer". It's always good to double-check your programs like this by running them multiple times as a quick way to confirm that they behave as intended.

Here's the full code of my program now (including the commented-out lines of code):

```
1  import random
2  f = open("petnames.txt", "r")
3  f_content = f.read()
4  f_content_list = f_content.split("\n")
5  print(random.choice(f_content_list))
```

Run
Reset

Finally, if I had multiple files in my folder, I could allow myself to pick a file from which to read in a list of names.

Here's how that would work:

```
1  import random
2  f_name = input('Type the file name: ')
3  f = open(f_name) # "r" omitted as it's the default
4  f_content = f.read()
5  f_content_list = f_content.split("\n")
6  print(random.choice(f_content_list))
```

Run
Reset

The only difference between this improvement and the previous program is that now I'm saving the `f_name` variable as the result of user-provided input. Once I have the `f_name` variable, I'm running the `open()` function on it and then proceeding with other steps as already explained.

✔ Completed Go to next item

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